HRGenius – Database Design

HCLTech Career Shaper Program | MySQL | Spring Data JPA / Hibernate

1. Database Overview

HRGenius uses MySQL as the primary relational database and H2 for testing. The design is based on the main entities in specification section 9.2 and the relationships in section 9.3. Eleven entities are listed in the original specification. Goal and the additional fields/endpoints described below are marked ASSUMPTION.

2. Entity and Column Definitions

User

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ORIGINAL (spec 9.2)
username	VARCHAR(100)	NOT NULL	UNIQUE	—	—	ASSUMPTION
password	VARCHAR(255)	NOT NULL	—	—	BCrypt hash	ORIGINAL (spec 9.2, 11)
role	VARCHAR(30)	NOT NULL	—	—	ADMIN/ HR_MANAGER/ MANAGER/ EMPLOYEE/ CANDIDATE; role constants ASSUMPTION C-3	ORIGINAL roles spec 5; constants ASSUMPTION
status	VARCHAR(20)	NOT NULL	—	—	ACTIVE/ INACTIVE (ASSUMPTION)	ORIGINAL field spec 9.2; values ASSUMPTION
employeeId	BIGINT	NULL	FK, UNIQUE	Employee(id)	Nullable for candidate accounts (ASSUMPTION)	ORIGINAL relationship spec 9.3; constraint ASSUMPTION
candidateId	BIGINT	NULL	FK, UNIQUE	Candidate(id)	ASSUMPTION C-6c	ASSUMPTION

Employee

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ORIGINAL spec 9.2
employeeCode	VARCHAR(20)	NOT NULL	UNIQUE	—	—	ASSUMPTION
name	VARCHAR(150)	NOT NULL	—	—	—	ORIGINAL field spec 9.2; nullability ASSUMPTION
email	VARCHAR(150)	NOT NULL	UNIQUE	—	—	ASSUMPTION
phone	VARCHAR(20)	NULL	—	—	—	ORIGINAL field spec 9.2; nullability ASSUMPTION
dateOfJoining	DATE	NOT NULL	—	—	—	ORIGINAL field

						spec 9.2; nullability ASSUMPTION
designation	VARCHAR(100)	NOT NULL	—	—	—	ORIGINAL field spec 9.2; nullability ASSUMPTION
status	VARCHAR(20)	NOT NULL	—	—	ACTIVE/ INACTIVE (ASSUMPTION)	ORIGINAL field spec 9.2; values ASSUMPTION
departmentId	BIGINT	NOT NULL	FK	Department(id)	—	ORIGINAL spec 9.2, 9.3
managerId	BIGINT	NULL	FK	Employee(id)	Nullable for top-level manager (ASSUMPTION)	ORIGINAL relationship spec 9.3; nullability ASSUMPTION

Department

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
name	VARCHAR(100)	NOT NULL	UNIQUE	—	—	ORIGINAL spec 9.2; uniqueness ASSUMPTION
description	VARCHAR(500)	NULL	—	—	—	ORIGINAL spec 9.2

JobOpening

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
title	VARCHAR(150)	NOT NULL	—	—	—	ORIGINAL spec 9.2
departmentId	BIGINT	NOT NULL	FK	Department(id)	—	ORIGINAL field spec 9.2; FK target ASSUMPTION
description	TEXT	NULL	—	—	—	ORIGINAL spec 9.2
vacancies	INT	NOT NULL	CHECK	—	> 0 (ASSUMPTION)	ORIGINAL field spec 9.2; check ASSUMPTION
status	VARCHAR(20)	NOT NULL	—	—	OPEN/ CLOSED (ASSUMPTION)	ORIGINAL field spec 9.2; values ASSUMPTION
postedDate	DATE	NOT NULL	—	—	—	ORIGINAL spec 9.2

Candidate

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
name	VARCHAR(150	NOT NULL	—	—	—	ORIGINAL

	)					spec 9.2
email	VARCHAR(150)	NOT NULL	—	—	—	ORIGINAL spec 9.2
phone	VARCHAR(20)	NULL	—	—	—	ORIGINAL spec 9.2
resume	VARCHAR(500)	NULL	—	—	Path or URL	ORIGINAL spec 9.2
jobOpeningId	BIGINT	NOT NULL	FK	JobOpening(id)	—	ORIGINAL spec 9.2, 9.3
applicationStatus	VARCHAR(30)	NOT NULL	—	—	APPLIED, SHORTLISTED, INTERVIEWED, SELECTED, REJECTED	ORIGINAL spec 9.1, 9.5

Interview

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
candidateId	BIGINT	NOT NULL	FK	Candidate(id)	—	ORIGINAL spec 9.2, 9.3
interviewerId	BIGINT	NOT NULL	FK	Employee(id)	Interviewer is Manager or HR (role mapping ASSUMPTION)	ORIGINAL field spec 9.2; FK target ASSUMPTION
scheduledDate	DATETIME	NOT NULL	—	—	—	ORIGINAL spec 9.2
feedback	TEXT	NULL	—	—	—	ORIGINAL spec 9.2
result	VARCHAR(30)	NULL	—	—	PENDING/ PASS/FAIL (ASSUMPTION)	ORIGINAL field spec 9.2; values ASSUMPTION

OnboardingTask

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
employeeId	BIGINT	NOT NULL	FK	Employee(id)	—	ORIGINAL spec 9.2
taskName	VARCHAR(150)	NOT NULL	—	—	—	ORIGINAL spec 9.2
status	VARCHAR(30)	NOT NULL	—	—	PENDING/ IN_PROGRESS/ COMPLETED (ASSUMPTION)	ORIGINAL field spec 9.2; values ASSUMPTION
dueDate	DATE	NULL	—	—	—	ORIGINAL spec 9.2

SalaryStructure

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
employeeId	BIGINT	NOT NULL	FK, UNIQUE	Employee(id)	One-to-one	ORIGINAL

					spec 9.3	spec 9.2, 9.3
basicPay	DECIMAL(12,2)	NOT NULL	CHECK	—	>= 0 (ASSUMPTION)	ORIGINAL field spec 9.2; check ASSUMPTION
hra	DECIMAL(12,2)	NOT NULL	—	—	—	ORIGINAL spec 9.2
allowances	DECIMAL(12,2)	NOT NULL	—	—	Other Allowances (ASSUMPTION C-5)	ORIGINAL field spec 9.2; meaning ASSUMPTION
deductions	DECIMAL(12,2)	NOT NULL	—	—	Other fixed deductions (ASSUMPTION C-4)	ORIGINAL field spec 9.2; meaning ASSUMPTION

Payroll

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
employeeId	BIGINT	NOT NULL	FK	Employee(id)	—	ORIGINAL spec 9.2, 9.3
month	TINYINT	NOT NULL	CHECK	—	1–12 (ASSUMPTION)	ORIGINAL field spec 9.2; check ASSUMPTION
year	SMALLINT	NOT NULL	—	—	—	ORIGINAL spec 9.2
grossPay	DECIMAL(12,2)	NOT NULL	—	—	—	ORIGINAL spec 9.2, 9.4
pf	DECIMAL(12,2)	NOT NULL	—	—	12% basic (ASSUMPTION C-4)	ASSUMPTION
tds	DECIMAL(12,2)	NOT NULL	—	—	Annual slab in Guidelines (ASSUMPTION C-4)	ASSUMPTION
otherDeductions	DECIMAL(12,2)	NOT NULL	—	—	SalaryStructure.deductions (ASSUMPTION C-4)	ASSUMPTION
totalDeductions	DECIMAL(12,2)	NOT NULL	—	—	PF + TDS + otherDeductions	ORIGINAL formula spec 9.4; component columns ASSUMPTION
netPay	DECIMAL(12,2)	NOT NULL	—	—	grossPay - totalDeductions	ORIGINAL spec 9.2, 9.4
status	VARCHAR(20)	NOT NULL	—	—	GENERATED/PAID (ASSUMPTION)	ORIGINAL field spec 9.2; values ASSUMPTION

PerformanceCycle

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
name	VARCHAR(100)	NOT NULL	—	—	—	ORIGINAL spec 9.2
startDate	DATE	NOT NULL	—	—	—	ORIGINAL

						spec 9.2
endDate	DATE	NOT NULL	—	—	Must be >= startDate (ASSUMPTION)	ORIGINAL field spec 9.2; check ASSUMPTION

PerformanceReview

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION: key implied by entity
employeeId	BIGINT	NOT NULL	FK	Employee(id)	—	ORIGINAL spec 9.2, 9.3
reviewerId	BIGINT	NOT NULL	FK	Employee(id)	Manager reviewer (role mapping ASSUMPTION)	ORIGINAL field spec 9.2; FK target ASSUMPTION
cycleId	BIGINT	NOT NULL	FK	PerformanceCycle(id)	—	ORIGINAL spec 9.2, 9.3
goals	TEXT	NULL	—	—	—	ORIGINAL spec 9.2
rating	TINYINT	NULL	CHECK	—	1–5 when set (ASSUMPTION)	ORIGINAL field spec 9.2; range ASSUMPTION
comments	TEXT	NULL	—	—	—	ORIGINAL spec 9.2
selfAssessment	TEXT	NULL	—	—	ASSUMPTION C-6b	ASSUMPTION
status	VARCHAR(20)	NOT NULL	—	—	PENDING/ FINAL (ASSUMPTION)	ASSUMPTION

Goal

Column	Data type	Nullability	Key	References	Constraint / Default	Source
id	BIGINT	NOT NULL	PK	—	AUTO_INCREMENT	ASSUMPTION C-6a
employeeId	BIGINT	NOT NULL	FK	Employee(id)	—	ASSUMPTION C-6a
cycleId	BIGINT	NOT NULL	FK	PerformanceCycle(id)	—	ASSUMPTION C-6a
title	VARCHAR(150)	NOT NULL	—	—	—	ASSUMPTION C-6a
description	TEXT	NULL	—	—	—	ASSUMPTION C-6a
status	VARCHAR(30)	NOT NULL	—	—	OPEN/ IN_PROGRESS/ COMPLETED (ASSUMPTION)	ASSUMPTION C-6a
setBy	BIGINT	NOT NULL	FK	Employee(id)	Manager who sets goal	ASSUMPTION C-6a

3. Relationships and Referential Actions

4. Entity Relationship Diagrams

The following diagrams present the database structure at three levels. Figure 1 is the consolidated ERD for the 11 entities explicitly listed in the original HRGenius specification (Section 9.2). Figures 2 and 3 are supporting module views that make the recruitment/onboarding and payroll/performance relationships easier to read.

Source alignment: entity names and baseline attributes follow Sections 9.2–9.4. Relationships explicitly stated in Section 9.3 are treated as specification relationships. Additional foreign keys, implementation types, uniqueness rules, and module workflow links shown as ASSUMPTION in the diagrams are proposed design details, not confirmed source requirements.

Baseline scope note: the specification stores goals in PerformanceReview.goals and lists 11 entities. The separate Goal entity retained in Section 2 is an optional ASSUMPTION extension; it is intentionally excluded from the baseline ERD.

Figure 1. Complete HRGenius HRMS ERD

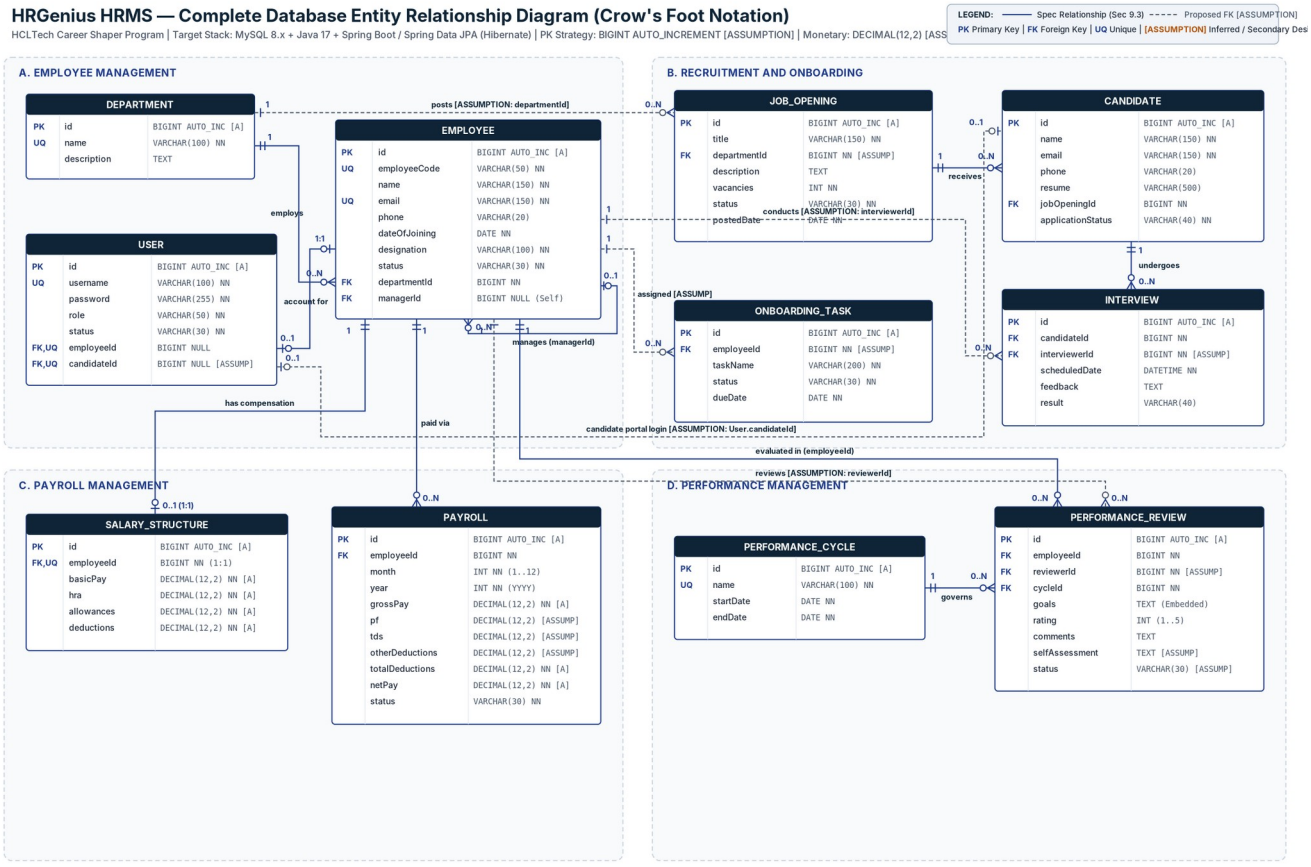


Figure 2. Recruitment and Onboarding ERD

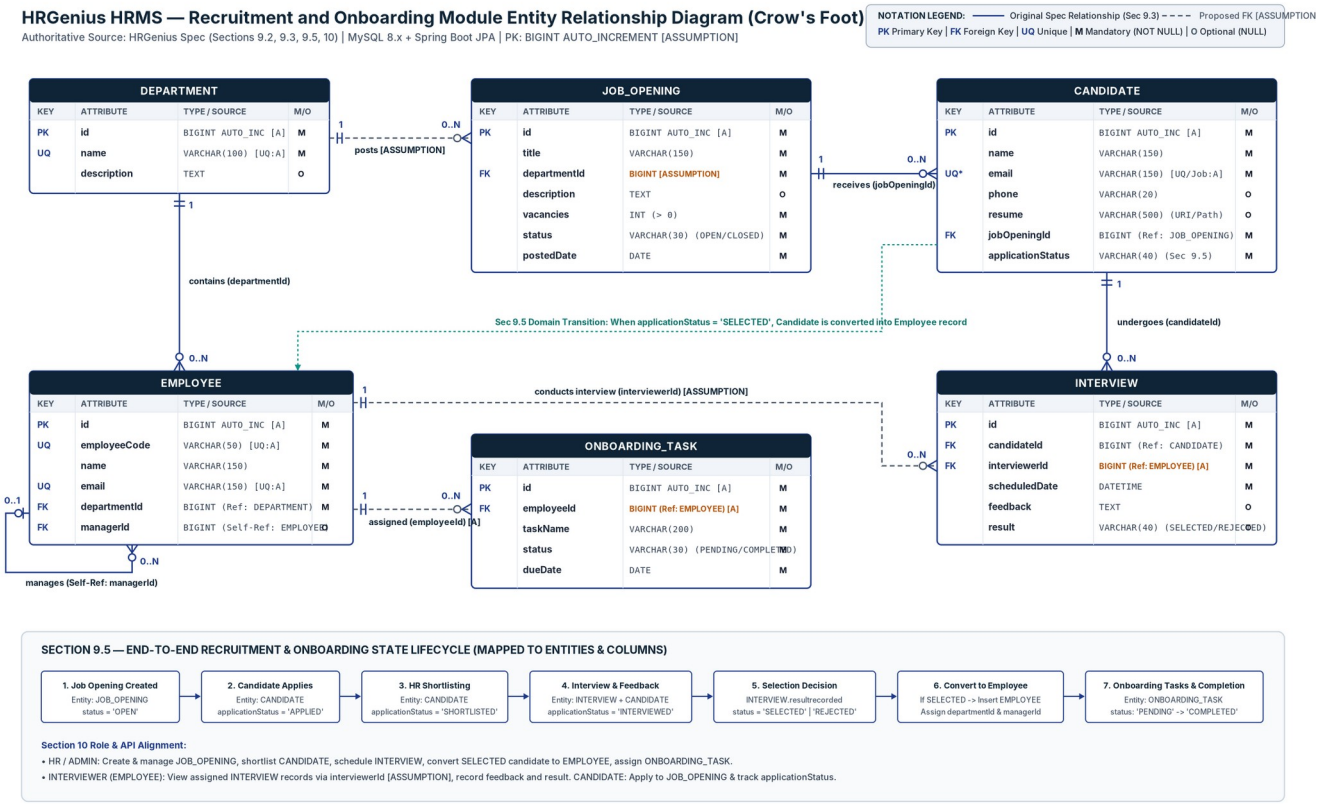


Figure 3. Payroll and Performance Management ERD

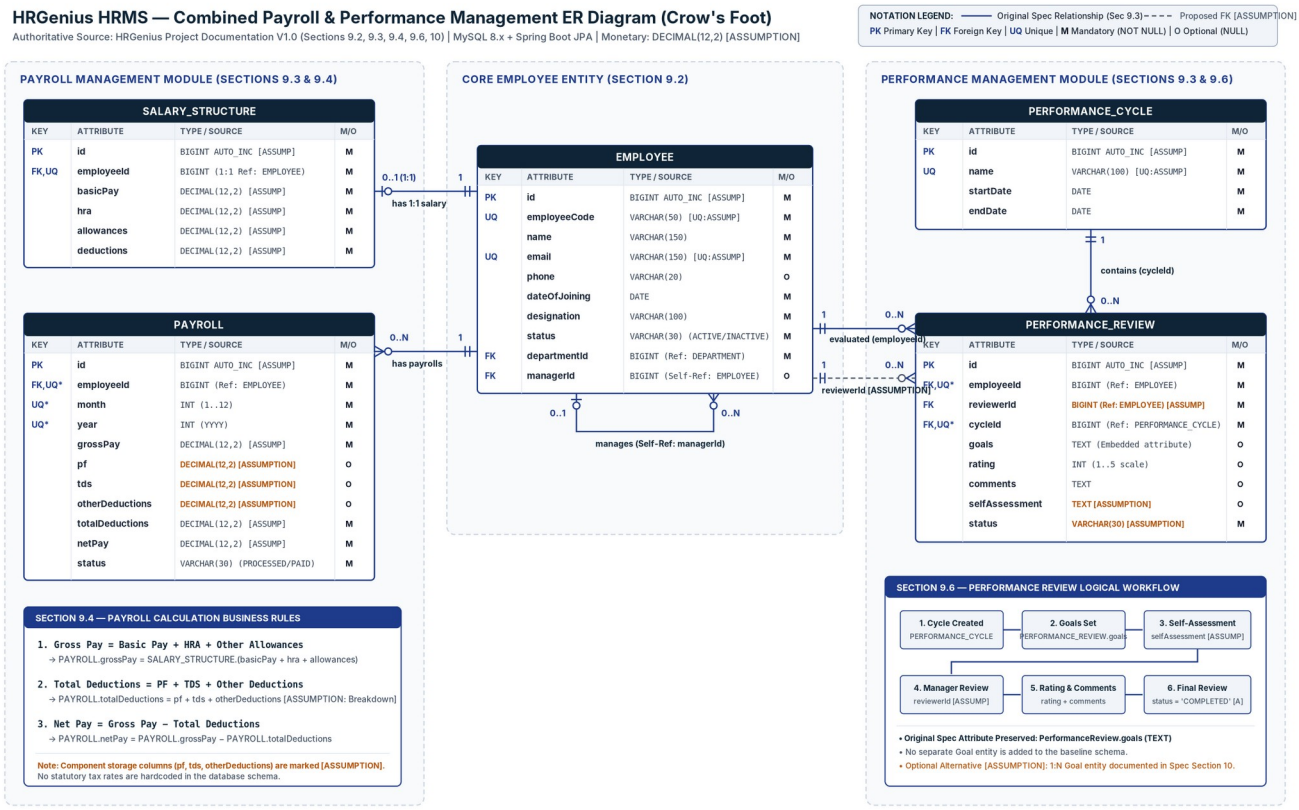


Diagram review note: the master diagram is the required consolidated view. The two module diagrams are supplementary views and do not introduce new baseline entities. Where a module diagram marks a composite uniqueness rule or an additional relationship with an asterisk/ASSUMPTION, treat it as a proposed implementation constraint pending confirmation.

5. Constraints and Indexes

- Primary key on id for every table.
- Unique constraints: Employee.employeeCode, Employee.email, User.username; one SalaryStructure per employee; one User per linked Employee; one Candidate per linked User (ASSUMPTION).
- Payroll uniqueness on (employeeId, month, year) prevents duplicate monthly payroll (ASSUMPTION).
- Foreign-key indexes on every FK column, including managerId, departmentId, jobOpeningId, candidateId, interviewerId, employeeId, reviewerId, cycleId, setBy, User.employeeId and User.candidateId.
- Composite index on Payroll(month, year); an additional index on Payroll(employeeId, year, month) supports employee payroll history (ASSUMPTION).
- CHECK vacancies > 0; month between 1 and 12; rating between 1 and 5 when rating is not null; cycle endDate >= startDate. CHECK enforcement depends on MySQL version.
- Use DECIMAL(12,2) for monetary values; do not use floating-point types for payroll.

6. Payroll Calculation Rules

Original formula (spec 9.4):

- Gross Pay = Basic Pay + HRA + Other Allowances.
- Total Deductions = PF + TDS + Other Deductions.
- Net Pay = Gross Pay − Total Deductions.

ASSUMPTION C-4: PF is 12% of Basic Pay. TDS uses the annualised gross-pay slab below; this is a simplified project calculation, not statutory tax advice. SalaryStructure.deductions represents other fixed deductions; Payroll stores the component deductions.

Annual gross pay	Illustrative annual TDS rate
Up to ₹3,00,000	0%
₹3,00,001–₹6,00,000	5%
₹6,00,001–₹10,00,000	10%
Above ₹10,00,000	15%

ASSUMPTION: TDS per month is annual slab tax divided by 12. Round each monetary component to two decimal places using HALF_UP. Mid-month joiners and exits are prorated by eligible calendar days in the month. The same proration applies to basic pay, HRA, allowances, PF and other fixed deductions; TDS is prorated by eligible days.

Worked example	Amount (₹)
Basic Pay	50000
HRA	20000
Other Allowances	5000
Gross Pay	75000
PF (12% of basic)	6000
Illustrative monthly TDS	625
Other Deductions	200
Total Deductions	6825
Net Pay	68175

Example 2 (illustrative): Basic ₹30,000 + HRA ₹12,000 + allowances ₹3,000 = gross ₹45,000. PF ₹3,600 + TDS ₹0 (if annualised gross is within the 0% slab) + other deductions ₹200 = total deductions ₹3,800. Net pay = ₹41,200.

7. Views

View	Purpose	Tables joined
EmployeeSummary	Employee profile with department and manager information	Employee, Department, Employee (manager)
RecruitmentPipeline	Job opening and candidate application stage overview	JobOpening, Candidate, Department
PayrollSummary	Monthly employee payroll components and totals	Payroll, Employee, SalaryStructure
PerformanceSummary	Employee ratings and cycle details	PerformanceReview, Employee, PerformanceCycle

HeadcountByDepartment	Employee count grouped by department and status	Employee, Department

Views are ASSUMPTION design extensions; the specification requires analytics outputs but does not prescribe SQL view objects.

8. Stored Procedures

Procedure	Purpose	Inputs	Validations / Transaction
RegisterEmployee	Create employee and related account where applicable	Employee details, department, account details	Validate required fields and unique employeeCode/email/username; transaction (ASSUMPTION)
ConvertCandidateToEmployee	Convert selected candidate; create employee, user and onboarding tasks	candidateId, employee details, user details, task details	Candidate must be Selected; perform atomically and rollback on failure (ASSUMPTION)
RunMonthlyPayroll	Generate payroll for target month	month, year	Validate month/year; calculate payroll; reject duplicates by employee/month/year (ASSUMPTION)
SubmitPerformanceReview	Save manager review	employeeId, reviewerId, cycleId, goals, rating, comments	Validate rating 1–5 and reviewer role (ASSUMPTION)
DeactivateEmployee	Deactivate employee without removing history	employeeId	Set status inactive; preserve payroll and review history (ASSUMPTION)

Stored procedures are ASSUMPTION extensions. The original specification describes application-layer modules and APIs, but does not require database stored procedures.

9. Analytics Query Definitions

Analytics	Definition	Source tables
Headcount by department	Count employees grouped by department; optionally separate ACTIVE and INACTIVE.	Employee, Department
Hiring trend by month	Count employees by month of dateOfJoining (ASSUMPTION A-07).	Employee
Attrition rate	Exits during period ÷ average headcount during period × 100 (ASSUMPTION A-07).	Employee; exit date field requires implementation agreement
Monthly payroll cost	Sum grossPay grouped by Payroll year and month (ASSUMPTION A-07).	Payroll
Rating distribution	Count reviews by rating grouped by performance cycle (ASSUMPTION A-07).	PerformanceReview, PerformanceCycle

10. Status Values

Field	Values	Source
User.status	ACTIVE, INACTIVE	ASSUMPTION
Employee.status	ACTIVE, INACTIVE	ACTIVE shown spec 10.9; INACTIVE ASSUMPTION
JobOpening.status	OPEN, CLOSED	ASSUMPTION
Candidate.applicationStatus	APPLIED, SHORTLISTED, INTERVIEWED, SELECTED, REJECTED	ORIGINAL spec 9.1, 9.5
Interview.result	PENDING, PASS, FAIL	ASSUMPTION
OnboardingTask.status	PENDING, IN_PROGRESS, COMPLETED	ASSUMPTION
Payroll.status	GENERATED, PAID	ASSUMPTION

PerformanceReview.status	PENDING, FINAL	ASSUMPTION
PerformanceCycle state	Derived from dates; ACTIVE/CLOSED if persisted	ASSUMPTION
Goal.status	OPEN, IN_PROGRESS, COMPLETED	ASSUMPTION

11. Assumptions Register

See HRGenius Guidelines section 15.10 for the consolidated register C-1 to C-7 and additional database, payroll and analytics assumptions. Design choices not explicitly stated in specification sections 9.2–9.4 are marked ASSUMPTION.